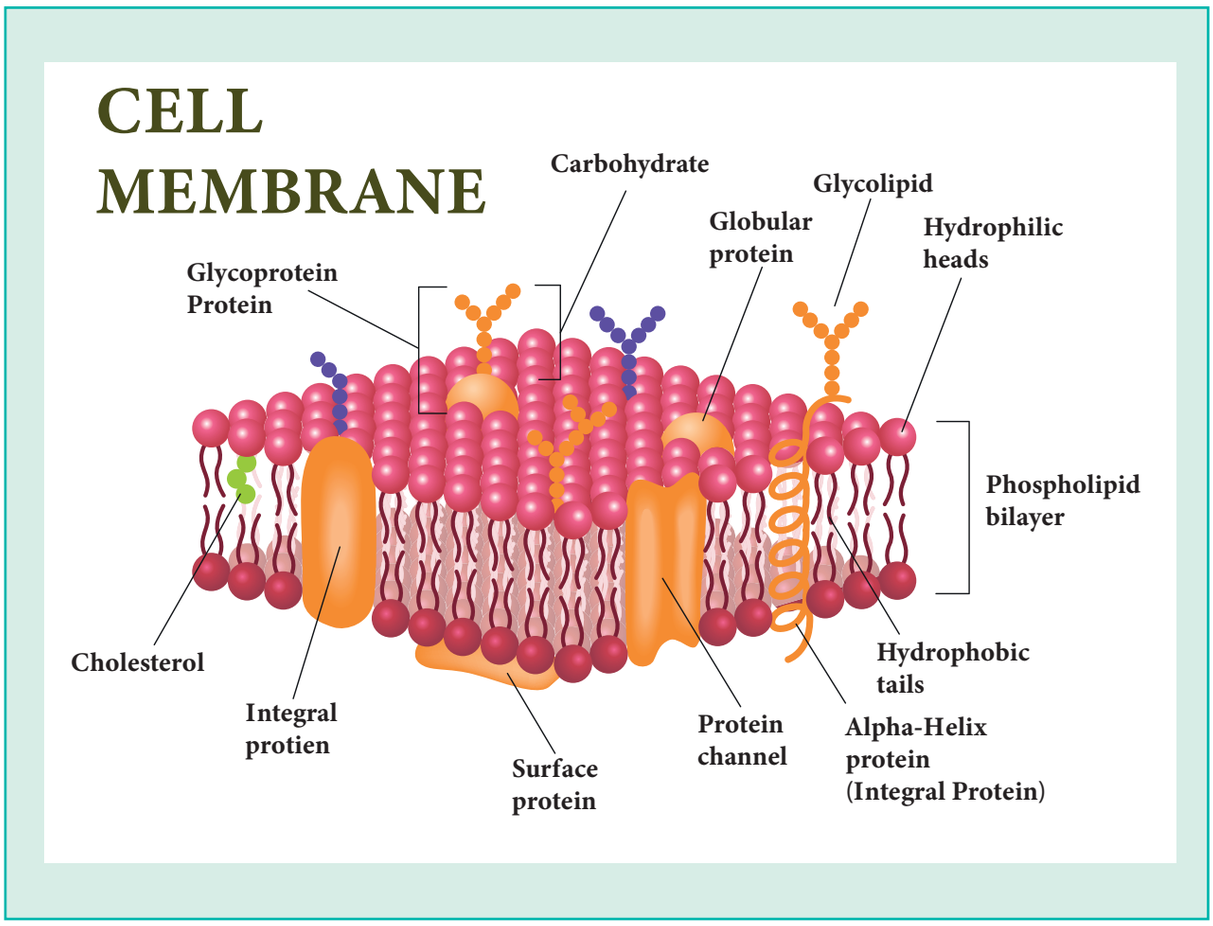
5.1 INTRODUCTION

Lipids (Greek: lipos-fat) are of great importance to the body as the chief concentrated storage form of energy, besides their role in cellular structure and various other biochemical functions. According to Bloor's criteria lipids are compounds having the following characteristics. They are insoluble in water and soluble in organic solvents (Chloroform, Benzene, Acetone etc.). Unlike polysaccharides, proteins and nucleic acids, lipids are not polymers. Further, lipids are mostly small molecules. Examples include fats, oils, waxes, sterols and fat-soluble vitamins.



5.1.1 Biological functions of Lipids

Lipids perform several important functions:

i. They are the concentrated fuel reserve of the body (triacylglycerols).

ii. Lipids are the constituents of biological membranes and regulate membrane permeability (phospholipids and cholesterol).

iii. They serve as a source of fat-soluble vitamins (A, D, E and K).

iv. Lipids are important as cellular metabolic regulators (steroid hormones and prostaglandins).

v. Lipids protect the internal organs, serve as insulating materials and give shape and smooth appearance to the body.

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